Digonto Biswas

Bhubaneswar, India

digontobiswas | digonto-biswas | https://digontobiswas.github.io

digontobiswas.kp.bd@gmail.com | +8801764348319 | Γ Google Scholar |

About me

I am an undergraduate Computer Science researcher specializing in Computer Vision. My research interests include medical imaging, autonomous vehicles, explainable AI for clinical support, and Vision Language Models (VLMs). My goal is to pursue a Ph.D. and develop robust AI systems.

Professional Experience

Research Intern July 2026 – Present

ELITE Research Lab LLC,Queens, NY, USA(Remote) elitelab.ai

• Research Intern at ELITE Research Lab LLC, contributing to research in artificial intelligence, computer vision, and autonomous vehicles through literature review, model development, experimental evaluation, and collaborative innovation.

Undergraduate Research Assistant & Co-Lead (AI/ML Systems) Aug 2025 – Present

KineTex Lab (KIIT Chapter), KIIT University, Bhubaneswar, India kinetex.in

• Conduct AI/ML-driven autonomous vehicle and biomedical research in medical imaging, multimodal learning, and XAI; develop deep learning and transformer models using large-scale clinical datasets, collaborate with PhD scholars on ongoing projects, contribute to publications, and mentor undergraduate researchers through structured research guidance.

Co-founder and Researcher May 2025 – Present

Biofolk, Remote biofolk.info

• Co-founded BioFOLK, an interdisciplinary research and knowledge-exchange platform, contributing to its vision and research direction, leading collaborative initiatives, and fostering interdisciplinary research culture, education, and global collaboration.

Education

2023 - present Bachelor of Technology in Computer Science and Engineering (CSE)
at **Kalinga Institute of Industrial Technology,** Bhubaneswar, India

Selected Publications

Journal

- Chakrabarty S., **Biswas, D.**, Ballove, T., Bandyopadhyay, A., Biswas, M. (2026). Efficient Deep Learning for Histopathology: A Comparative Analysis of YOLO26 and CNNs for Oral Cancer Classification. Available Preprint at SSRN

Conference Papers

- Halder, M., Sinha, P., **Biswas, D.**, Yang, T., & Rathore, R. S. (2025, March). Deployment of Machine Learning Model for Predictive Analysis of Rising Crime Trends. In *Doctoral Symposium on Computational Intelligence* (pp. 419–433). Singapore: Springer Nature Singapore. DOI: 10.1007/978-981-96-8104-4_27
- A. Das, P. K. Pattnaik, U. Sarkar, S. Mohajon, Turjya, **D. Biswas**, A. Bandyopadhyay, "Enhanced Autism Spectrum Disorder Detection Using Deep Multimodal EEG–Facial Image Fusion," *16th ICCCNT 2025*, Manuscript Presented.
- S. Chakrabarty, P. Bishwas, R. Chatterjee, T. Bandyopadhyay, **D. Biswas**, B. Howlader, "Soundscapes in Spectrograms: Pioneering Multilabel Classification for South Asian Sounds," *16th International Conference on Computing, Communication and Networking Technologies (ICCCNT 2025)*, Manuscript Presented. Preprint Available
- **D. Biswas**, B. Howlader, T. Ballove, M. R. B. Shahriar, M. D. Hossain, and A. Bandyopadhyay, "Computational Mapping of Human Host Proteins Associated with HMPV Using Protein–Protein Interaction Networks," 4th Doctoral Symposium on Human Centered Computing (Human-2026), Manuscript Accepted.
- P. Chakraborty, A. Bandyopadhyay, **D. Biswas**, "Interpretable Semantic–Depth Risk Estimation for Region-Aware Decision in Autonomous Driving," *2026 IEEE Region 10 Conference (TENCON,Intelligent Systems for a Resilient and Sustainable Society, Bali Resort, Jimbaran, BALI, indonesia), Manuscript Accepted.*

Book Chapters

- Jana, M.K., Mukherjee, P., Swarup, V., Ray, D., **Biswas, D.**, Howlader, B., Baidya, A., Das, S., Chiang, M.H., Jagadeb, M. and Qazi, S., 2026. AI in cancer drug development and immunotherapy. In Artificial Intelligence in Precision Drug Design, Volume 2 (pp. 93-123). Academic Press. DOI

- **D. Biswas**, S. Roy, P. Majumder, T. Ballove, A. Londhe, V. Swarup, M. Narayan, "Cloud and HPC Platforms for Molecular Docking and Simulation," in *Recent Advances in Molecular Docking: Methods, Tools, and Applications*, Academic Press (Elsevier), 2026. DOI

UNDER REVIEW

- **D Biswas**, P. Gupta, T. Ballove, S. Roy, H. Mohapatra, and A. B. Prangon, *"Quantum-driven AI agents in healthcare and genomics," in Quantum Computing and AI Agents for Intelligent Systems: Next-Generation Autonomy,* CRC Press, Taylor & Francis Group, 2026.

Ongoing Research

Transformer-Based Radiomic Token Embedding for Prostate Cancer Classification
Developed a transformer-based deep learning model for prostate cancer classification using radiomic features from multi-parametric MRI. Leveraged self-attention to better capture feature relationships and improve interpretability, contributing to higher diagnostic accuracy and more clinically meaningful insights.

Antibiotic Effectiveness and Treatment Outcome Analysis in Sepsis
Built a time-aware model to track how patients respond to antibiotics by combining structured chest CT findings with lab and microbiology data. Modeled disease progression over time to predict treatment outcomes, helping with earlier detection of treatment failure and more informed clinical decision-making.

Academic Projects

XV6-Null-Pointer-Dereference-in-Two-Level-Page-Tables Link to Demo
This project explores how XV6 (a teaching OS for RISC-V) handles null pointer dereferences in its two-level page table system. We analyze key kernel mechanisms, including: Address space initialization via exec() Page table copying during fork() Kernel page fault handling for null pointer dereferences

PCA and Regression Study of Nitrate–Boron in NWMP Waters Link to Demo
PCA-based projection and regression analyzed NWMP water quality data for Nitrate and Boron. July-August data trained models, while September data tested them. PCA maintained 97 percent variance. Linear and Logistic Regression models were compared, with Logistic performing better.

Awards and Achievements

Achievements

- **Competitive Programming:** Achieved Pupil rank on Codeforces with a peak rating of 1208 in Round 1039 (Div. 2), demonstrating strong problem-solving skills.
- **KUISP SAARC Scholarship (KIIT):** Awarded the KUISP SAARC Scholarship covering 60% of tuition, accommodation, meals, and examination fees for the B.Tech CSE program (2023–2024) at KIIT, India.
- **Walton International Rating Chess Tournament 2023:** Secured 2nd position with 5/7 points at the Walton International Rating Chess Tournament in Dhaka, Bangladesh.
- **Best Reader Award – World Literature Center:** Received the Best Reader Award (2014) for outstanding performance in the Learning Development Program (LDP) under SEQAEP, Ministry of Education, Bangladesh.

Extra Curricular Activities and Volunteering Work

- **Head of the ICT Cell,** Sheikh Russel National Children and Adolescent Council
- **Cub Leader,** Second Kalapara Upazila Cub Campuri 2011, Bangladesh Scouts
- **Unit Leader,** Patuakhali District Cub Campuri 2012

Skills

- **Programming Languages:** Python, C++, Java, HTML, CSS, JavaScript, PHP, Django, Odoo
- **ML Platforms:** PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV.
- **Database Management Systems:** Experienced with Oracle and MySQL
- **Tools & Technologies:** Git & GitHub, LaTeX
- **Additional Skills:** Developing Machine Learning-based Solutions, Cisco, MikroTik, AWS

REFERENCES

◇ **Dr. Anjan Bandyopadhyay**
Nodal Officer, Visvesvaraya PhD Scheme
(MeitY, Govt. of India)
Associate Professor
Dept. of Computer Engineering
KIIT University, India
Email: anjan.bandyopadhyayfcs@kiit.ac.in
Phone: +91-9040820056

◇ **Dr. Mainak Biswas**
Associate Professor
Dept. of Computer Engineering
KIIT University, India
Email: mainak.biswasfcs@kiit.ac.in
Phone: +91-7721827792